

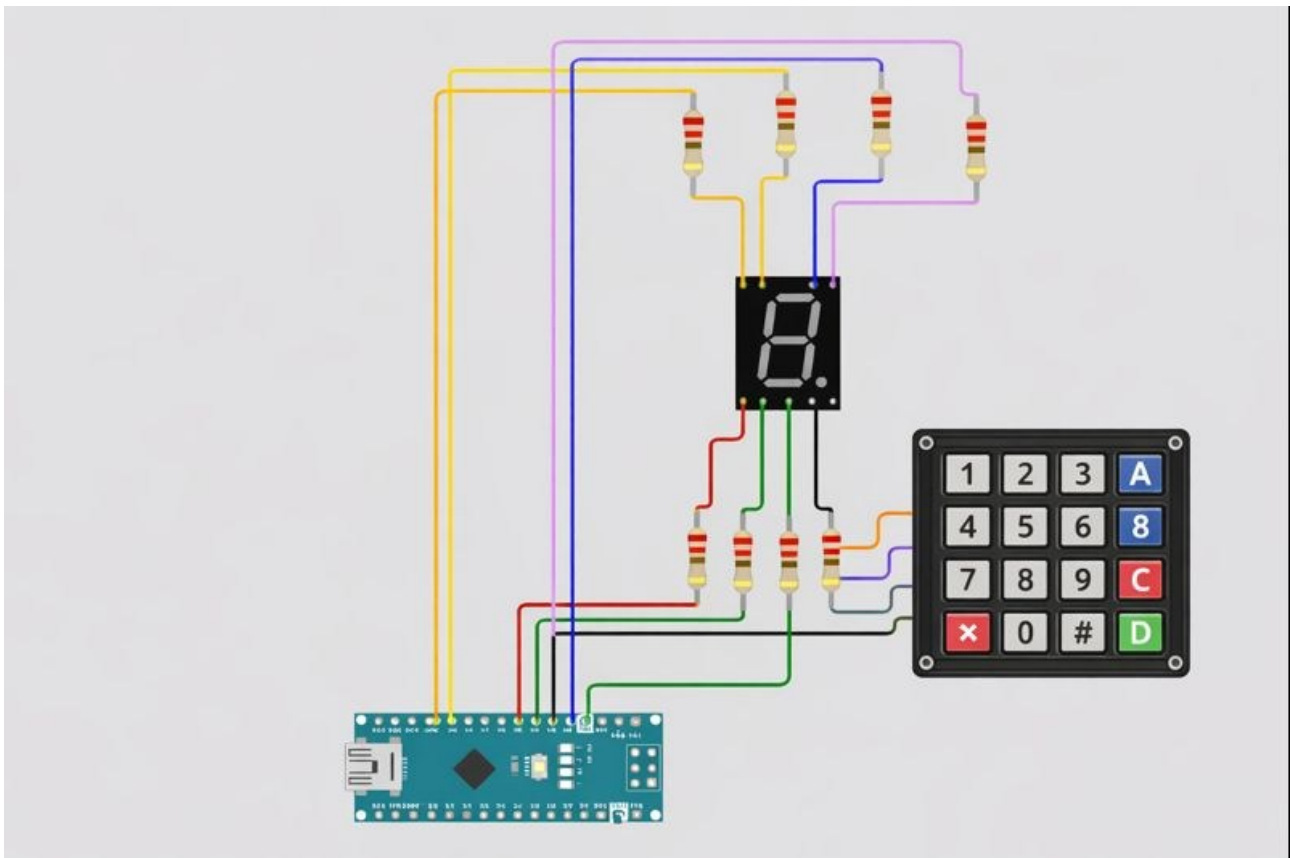
Keypad-Synchronized Multi-Digit 7-Segment Display Using Arduino Nano

This project uses an Arduino Nano to read keypad inputs and display the corresponding value in real time on a 7-segment display..

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Components and supplies

- 1 Arduino Nano Matter
- 1 Button Keypad 4x4 module
- 4 Resistor 220 ohms
- 1 7 Segment Display-Common Anode



Project description

This project demonstrates how to interface a 4×4 matrix keypad with an Arduino Nano to control a 7-segment display. Each key pressed on the keypad is processed by the Arduino and displayed in real time on the 7-segment display, creating a simple and responsive human–machine interface.

The project focuses on understanding digital input scanning, keypad matrix decoding, and output control of a 7-segment display using current-limiting resistors. It serves as a foundation for applications such as access systems, counters, menu selection interfaces, and basic embedded user input systems.

1. The 4×4 keypad is connected to the Arduino Nano using row and column pins.
2. The Arduino continuously scans the keypad matrix to detect which key is pressed.
3. When a key is detected, the Arduino maps the key value to its corresponding 7-segment pattern.
4. The 7-segment display is driven directly from the Arduino's digital pins through resistors to protect the LEDs.
5. Each keypad press is synchronized with the display, ensuring that the selected number or character appears instantly on the 7-segment display. Experimente: Atualize o Sumário.

Code

```
#include <Keypad.h>

/* ===== Keypad setup ===== */
const byte ROWS = 4;
const byte COLS = 4;

char keys[ROWS][COLS] = {
  {'1', '2', '3', 'A'},
  {'4', '5', '6', 'B'},
  {'7', '8', '9', 'C'},
  {'*', '0', '#', 'D'}
};
```

```

byte rowPins[ROWS] = {9, 8, 7, 6};    // Connect to keypad
ROW pins
byte colPins[COLS] = {5, 4, 3, 2};    // Connect to keypad
COL pins

Keypad keypad = Keypad(makeKeymap(keys), rowPins, colPins,
ROWS, COLS);

/* ===== 7-segment pins (a-g) ===== */
int segA = 10;
int segB = 11;
int segC = 12;
int segD = A0;
int segE = A1;
int segF = A2;
int segG = A3;

/* ===== Segment patterns for numbers 0-9 ===== */
/* Common Cathode (HIGH = ON) */
byte digits[10][7] = {
    {1,1,1,1,1,1,0}, // 0
    {0,1,1,0,0,0,0}, // 1
    {1,1,0,1,1,0,1}, // 2
    {1,1,1,1,0,0,1}, // 3
    {0,1,1,0,0,1,1}, // 4
    {1,0,1,1,0,1,1}, // 5
    {1,0,1,1,1,1,1}, // 6
    {1,1,1,0,0,0,0}, // 7
    {1,1,1,1,1,1,1}, // 8
    {1,1,1,1,0,1,1}  // 9
};

int segmentPins[7] = {segA, segB, segC, segD, segE, segF,
segG};

```

```

void setup() {
    for (int i = 0; i < 7; i++) {
        pinMode(segmentPins[i], OUTPUT);
        digitalWrite(segmentPins[i], LOW);
    }
}

void loop() {
    char key = keypad.getKey();

    if (key >= '0' && key <= '9') {
        displayDigit(key - '0');
    }
}

/* ===== Display function ===== */
void displayDigit(int num) {
    for (int i = 0; i < 7; i++) {
        digitalWrite(segmentPins[i], digits[num][i]);
    }
}

```